

SunScan Homeowner Calculator Brief

1. Project Overview

SunScan is being developed as a solar and battery platform for installers, similar in purpose to OpenSolar, Easy PV and other PV design tools, but with a stronger focus on ease of use, homeowner-friendly recommendations, survey proposal generation and accurate kit list creation.

The full SunScan platform will eventually help installers design systems, create customer proposals and generate cost-efficient kit lists for the whole job, including panels, inverters, batteries, mounting equipment, rails, hooks and other installation materials.

Before the full platform is built, SunScan will launch a slimmed-down homeowner-facing calculator. This will be used by installers as a lead generation tool.

HSEnergy Residential will be used as the first test-case installer.

2. Purpose of the Slimmed-Down Calculator

The calculator should allow homeowners to quickly understand what solar and battery system could suit their home, roof space, energy usage, tariff, budget and objectives.

Unlike a basic questionnaire, the calculator should include roof mapping and panel layout functionality. The homeowner enters their address, SunScan uses Google Earth or similar mapping data to locate the property, the homeowner confirms the correct property, and SunScan models the roof to estimate how many solar panels could fit.

The calculator should then present three indicative system options. These should be complete solar and battery packages.

Each option should compare:

- Number of solar panels
- Solar panel type
- Inverter type
- Battery type and capacity
- System quality level
- Estimated generation
- Estimated savings
- Indicative installed price
- Estimated payback period
- Sentinel uplift as a separate optimisation layer

The main commercial objective is to generate qualified homeowner leads by encouraging the user to request a detailed report or book a free remote or onsite survey.

3. Homeowner Journey Summary

The journey should follow this structure:

1. Homeowner lands on the calculator page.
 2. They enter their address.
 3. SunScan finds the property using Google Earth or similar mapping data.
 4. The homeowner confirms the correct property.
 5. SunScan models the roof and estimates possible panel layout and maximum panel capacity.
 6. The homeowner provides energy usage, tariff, budget and motivation details.
 7. SunScan generates three complete system options.
 8. The options compare budget, balanced and recommended system scenarios.
 9. Sentinel is shown as a separate uplift layer across all three options.
 10. The homeowner can request a detailed report by submitting contact details.
 11. The homeowner can also book a free remote or onsite survey earlier in the journey if they prefer.
 12. The installer receives the lead, including property details, usage inputs, system options, Sentinel uplift, lead score and requested next step.
-

4. Landing Page

Objective

Encourage the homeowner to start the calculator.

Suggested headline

"Find out what solar and battery system could work for your home."

Supporting message

"Enter your address and answer a few simple questions. SunScan will model your roof, estimate how many panels could fit and show you three system options based on your budget, energy usage and potential savings."

Primary CTA

"Start My Estimate"

Secondary CTA

"Book a Free Survey"

The survey CTA should allow users to skip the calculator if they already want to speak to the installer.

5. Step 1: Address and Property Confirmation

Question

What is your address?

Required input

- Address lookup field
- Postcode search
- Manual address entry if lookup fails

Process

After the address is entered, SunScan should locate the property using Google Earth or similar mapping data.

The homeowner should then be shown the property and asked to confirm it.

Confirmation question

Is this the correct property?

Options:

- Yes, this is correct
 - No, let me search again
 - I need to adjust the property location
-

6. Step 2: Roof Mapping and Panel Layout

Once the property is confirmed, SunScan should model the roof and estimate:

- Usable roof areas
- Roof orientation
- Likely shading risks where possible
- Potential panel layout
- Maximum number of panels
- Estimated system size in kWp

Homeowner-facing wording

"We've estimated how many panels may fit on your roof. This is an initial guide and will be confirmed by survey."

Important note

The calculator should not ask the homeowner to manually estimate roof size, roof orientation or shading unless the mapping process cannot determine it accurately.

If mapping confidence is low, the homeowner can be asked simple fallback questions.

7. Fallback Roof Questions

Only use these if the mapping tool cannot confidently assess the roof.

Roof size

How much usable roof space do you think you have?

Options:

- Small roof
- Medium roof
- Large roof
- I'm not sure

Roof direction

Which direction does your main roof face?

Options:

- South
- South East
- South West
- East
- West
- East and West split roof
- North
- I'm not sure

Shading

Is your roof affected by shading?

Options:

- No obvious shading
- Some shading from trees
- Some shading from nearby buildings
- Heavy shading
- I'm not sure

If heavy shading is selected, the results should clearly state that a survey is strongly recommended.

8. Step 3: Property and Ownership Details

Property type

What type of property do you live in?

Options:

- Detached house
- Semi-detached house
- Terraced house
- Bungalow
- Flat or apartment
- Other

Ownership status

Do you own your home?

Options:

- Yes, I own the property
- Yes, with a mortgage
- I rent privately
- I live in social housing
- I am a landlord
- Other

If renter or social housing is selected, continue the journey but flag that permission may be required.

9. Step 4: Electricity Usage

The ideal route is for the homeowner to upload a recent electricity bill so SunScan can extract the relevant information.

Primary question

Do you have a recent electricity bill available?

Options:

- Yes, upload a bill
- Yes, I'll enter the details manually
- No, I'll provide an estimate

Option A: Bill upload

Allow the homeowner to upload:

- Photo
- PDF
- Screenshot

SunScan should aim to extract:

- Annual electricity usage in kWh
- Current tariff
- Current unit rate
- Standing charge
- Supplier
- Export tariff if applicable

Option B: Manual usage entry

Ask:

Roughly how much electricity do you use each year?

Input:

- Annual kWh field

Optional helper text:

"You can usually find this on your electricity bill or energy supplier app."

Option C: Estimated usage

If the homeowner does not know their annual kWh, ask:

What is your typical monthly electricity bill?

Input:

- Monthly cost in £

Optional:

What is your average unit rate?

- I know my unit rate
- I'm not sure

Last-resort fallback

Only ask household size if the homeowner cannot provide a bill, kWh figure or bill cost.

How many people live in the property?

Options:

- 1
- 2
- 3
- 4
- 5+
- Prefer not to say

The results should state that recommendations are less accurate if based only on household size.

10. Step 5: Energy Lifestyle

Question

Which of these apply to your home?

Multiple choice:

- Someone is usually home during the day
- Most energy is used in the evening
- We have an electric vehicle
- We plan to get an electric vehicle
- We have a heat pump
- We plan to get a heat pump
- We have high daytime usage
- None of these
- Not sure

This helps determine battery size, self-consumption potential, tariff suitability and future-proofing.

11. Step 6: Tariff Type

Question

What type of electricity tariff are you on?

Options:

- Standard fixed or variable tariff
- Economy 7 or off-peak tariff
- Smart time-of-use tariff
- Import and export tariff
- I already export solar energy
- I'm not sure

If a bill has been uploaded, SunScan should use the bill data where possible instead of relying only on this answer.

12. Step 7: Existing Solar or Battery

Question

Do you already have solar panels or battery storage?

Options:

- No, I'm looking for a new system
- Yes, I already have solar panels
- Yes, I already have battery storage
- Yes, I have solar and battery storage
- Not sure

Existing system leads should be tagged separately as upgrade, retrofit or optimisation opportunities.

13. Step 8: Motivation and Budget

Main motivation

What is your main reason for considering solar and battery storage?

Options:

- Reduce electricity bills
- Protect against rising energy prices
- Make better use of cheap electricity tariffs
- Earn from exporting energy
- Become more energy independent
- Reduce carbon emissions
- Add value to my home
- I'm just exploring options

Budget range

What budget range are you considering?

Options:

- Under £6,000
- £6,000 to £8,000
- £8,000 to £10,000
- £10,000 to £12,000
- £12,000 to £15,000
- £15,000+
- I'm not sure yet

Finance interest

Would you be interested in finance options?

Options:

- Yes
- No
- Maybe
- I'd like to understand the options

14. Early Survey Option

The homeowner should not have to complete the full journey before requesting help.

At key points in the calculator, include a secondary CTA:

"Prefer expert help? Book a free survey instead."

This should be available:

- On the landing page
- After property confirmation
- After roof modelling
- Before the results page
- On the results page

Survey options

- Free remote survey
- Free onsite survey
- Let the installer recommend the best option

For the HSEnergy test case, both remote and onsite surveys should be available.

For future installers, survey options should be configurable.

15. Results Page

The results page should provide a helpful but concise overview. It should not be positioned as a final quote.

Suggested wording

“Based on your property, roof layout, energy usage and budget, your home may be suitable for solar and battery storage. We’ve created three indicative options to help you compare cost, performance and long-term value.”

Disclaimer

“This is an indicative estimate only. A final recommendation depends on a roof survey, electrical checks, shading assessment, confirmed usage and tariff details.”

16. Three System Options

SunScan should show three complete system scenarios. These should not just be “cheap, medium and expensive.” They should demonstrate how system size, component quality, battery capacity and Sentinel optimisation affect savings, earnings and payback.

Option 1: Budget-Fit System

This option is shaped around the homeowner’s stated budget.

Example:

The roof may have space for 20 panels, but the budget may only allow 10 panels and a smaller battery.

Positioning:

“Designed to fit closer to your stated budget, while still giving you a practical entry point into solar and battery storage.”

Typical logic:

- Fewer panels than maximum roof capacity
- Budget or standard solar panels
- Standard inverter
- Smaller or entry-level battery
- Lower upfront cost
- Longer payback period
- Sentinel uplift shown separately

Option 2: Better Value System

This option may exceed the original budget slightly but should improve long-term value.

Example:

A 16-panel system may cost more than the budget-fit option, but it may generate more electricity, make better use of battery storage and allow Sentinel to optimise more value from the system.

Positioning:

“Designed to improve long-term savings and payback by increasing system performance while keeping cost under control.”

Typical logic:

- More panels than the budget-fit system
- Good-quality solar panels
- Reliable hybrid inverter
- Better-sized battery
- Stronger self-consumption potential
- Better payback period
- Sentinel uplift shown separately

Option 3: Recommended System

This is the best overall recommendation based on roof potential, energy usage, tariff opportunity, battery sizing, Sentinel uplift and long-term return.

Example:

The roof may support 20 panels. With a stronger battery and Sentinel optimisation, the system may cost more upfront but deliver a much stronger return, reducing the payback period.

Positioning:

“Recommended for maximising long-term savings, system performance, optimisation and return on investment.”

Typical logic:

- Uses more of the available roof capacity
 - Higher-quality or higher-efficiency panels
 - Higher-spec hybrid inverter
 - Larger or more advanced battery
 - Stronger suitability for smart tariffs and export value
 - Strongest Sentinel uplift potential
 - Best projected long-term value
-

17. Results Comparison Table

The results page should show a simple comparison table.

Suggested columns:

- Option name
- Number of panels
- Solar panel type
- Inverter type
- Battery type
- Battery capacity
- Estimated generation
- Indicative installed price
- Estimated annual saving
- Estimated export / earning potential
- Estimated payback period
- Sentinel uplift
- Best suited to
- Recommended next step

The homeowner-facing labels should remain simple:

- Budget-fit
 - Better value
 - Recommended
-

18. Sentinel Optimisation Layer

Sentinel should be shown as a separate uplift layer across all three options.

It should not be hidden inside the package. The homeowner should be able to understand the estimated performance before and after Sentinel.

Simple homeowner explanation

“Your solar panels, inverter and battery create and store the energy. Sentinel helps the system make smarter decisions about when to charge, store, use or export that energy.”

Sentinel should show uplift through:

- System monitoring
- Fault or underperformance alerts
- Smarter battery charging
- Better use of tariff data
- Weather-aware optimisation
- Improved self-consumption
- Export and earning opportunities where suitable
- Greater confidence that the system is working as it should

Suggested display

For each system option, show:

Without Sentinel:

- Estimated annual savings: £X
- Estimated export value: £X
- Estimated payback: X years

With Sentinel:

- Estimated annual savings: £X
- Estimated export / earning potential: £X
- Estimated optimisation uplift: +X%
- Estimated payback: X years
- Monitoring and protection: Included

All figures should be labelled as indicative.

19. Recommendation Logic

The calculator should highlight one recommended option based on the combined answers and roof modelling.

Key recommendation factors

- Maximum roof panel capacity
- Customer budget
- Annual electricity usage
- Tariff type
- EV or heat pump plans
- Daytime vs evening usage
- Battery size suitability
- Sentinel optimisation potential
- Payback period
- Long-term ROI

Example logic

If the customer has roof space for 20 panels but only enters a budget suitable for 10 panels, show the 10-panel option as the Budget-Fit System.

Then show how increasing to 16 panels and a better battery may improve payback.

Then show the full recommended system using more roof capacity, better battery storage and Sentinel uplift if this produces the strongest long-term return.

Example homeowner wording

"Your budget suggests the Budget-Fit System may be the most affordable starting point. However, based on your roof capacity and energy usage, the Recommended System is likely to deliver stronger long-term value and a shorter estimated payback period."

20. Lead Capture and Detailed Report

After the results summary, the homeowner should be encouraged to request a detailed report.

CTA

"Get My Detailed Report"

Supporting text

"Your detailed report will show how different panel layouts, system sizes, panels, inverters, batteries and Sentinel optimisation could affect your savings, payback and long-term return."

Required fields

- First name
- Last name
- Email address
- Phone number
- Address
- Postcode

Optional fields

- Preferred contact method
- Best time to contact
- Additional comments

Consent checkbox

"I agree to be contacted about my solar and battery estimate."

21. Final Confirmation

If report requested

"Thanks, your report request has been received. A member of the team will review your details and send your personalised solar and battery estimate."

If survey requested

"Thanks, your survey request has been received. A member of the team will contact you to confirm the next step."

Optional note:

"To make your estimate more accurate, please have a recent electricity bill or annual electricity usage figure available."

22. Installer Lead Notification

The installer should receive a structured lead notification.

Suggested fields:

- Lead source: SunScan Calculator
 - Installer name
 - Customer name
 - Email
 - Phone
 - Address
 - Postcode
 - Property type
 - Ownership status
 - Roof model confidence
 - Estimated maximum panel count
 - Estimated system size potential
 - Electricity usage source: bill upload / manual kWh / bill cost / household estimate
 - Annual electricity usage
 - Monthly electricity cost
 - Tariff type
 - Existing solar or battery
 - EV / heat pump status
 - Main motivation
 - Budget range
 - Finance interest
 - Three options shown
 - Recommended option
 - Sentinel uplift shown
 - Estimated payback periods
 - Detailed report requested
 - Survey requested
 - Survey type requested
 - Lead score
 - Date and time submitted
-

23. Lead Scoring

The system should apply a simple lead score.

Hot lead

Likely criteria:

- Owns property
- Confirmed address
- Good roof model confidence
- Strong roof capacity
- Medium to high electricity usage

- Budget above £8,000
- Uploaded bill or provided kWh usage
- Requested survey
- Provided phone number

Warm lead

Likely criteria:

- Owns property
- Usable roof capacity
- Budget unclear or mid-range
- Provided partial usage information
- Requested detailed report but not survey

Nurture lead

Likely criteria:

- Renter
- Flat or apartment
- Low budget
- Poor or uncertain roof suitability
- Usage based only on household estimate
- Report requested only

24. Installer Configuration Requirements

The calculator should be built so different installers can configure:

- Solar panel options
- Inverter options
- Battery options
- Sentinel availability
- Package names
- Pricing rules
- Performance assumptions
- Warranty information
- Survey options
- Coverage areas
- Finance options
- Lead notification email
- CRM integration route
- Minimum and maximum system sizes
- Preferred product combinations
- Margin assumptions

Each product should ideally be tagged by tier:

- Budget
- Standard
- Premium

This allows SunScan to build package options automatically.

25. Key Design Principles

The calculator should be:

- Simple for homeowners to complete
- Built around address-based roof modelling
- Clear that results are indicative, not a final quote
- Helpful before asking for contact details
- Focused on value, not just cheapest price
- Able to show budget-fit vs better-value vs recommended systems
- Able to show the impact of Sentinel as a separate uplift layer
- Flexible enough to be configured for different installers
- Built around lead generation as the primary objective

The key message for homeowners should be:

“Your cheapest option may not always give you the best long-term return. A larger or better-designed system can cost more upfront but may deliver stronger savings, better payback and greater long-term value.”

26. Future Full Platform Functionality

The future full SunScan platform may include:

- Full installer design dashboard
- Advanced roof mapping and panel layout
- Shading analysis
- System design
- Battery sizing
- Inverter compatibility checks
- Full kit list generation
- Rails, hooks and mounting recommendations
- Cost-efficient ordering suggestions
- Supplier pricing integration
- Proposal generation
- Installer margin settings
- CRM integration

- Remote survey upload area
- Finance integration
- Tariff and optimisation modelling
- Sentinel compatibility logic